Customer Churn Prediction & Risk Segmentation Report

Project Title

Customer Churn Prediction & Risk Segmentation Dashboard

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1. Introduction

Customer churn is one of the biggest challenges faced by subscription-based businesses. Losing customers directly affects company revenue, customer growth, and long-term business stability. The purpose of this project was to build a machine learning pipeline capable of predicting whether a telecom customer is likely to churn and identifying different customer risk levels.

This project focused on performing exploratory data analysis, data preprocessing, feature engineering, model training, customer risk segmentation, and business insight generation using a real-world telecom churn dataset.

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2. Dataset Overview

The dataset used in this project was the Telco Customer Churn dataset from Kaggle. It contains customer demographic information, billing details, internet service information, contract details, and churn status.

Dataset Information:

- Total Rows: 7,043
- Total Columns: 21
- Target Variable: Churn (Yes/No)

Important features in the dataset included:

- Tenure
- MonthlyCharges
- TotalCharges
- Contract Type
- Internet Service
- Payment Method
- Online Security
- Tech Support

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3. Data Preprocessing & Feature Engineering

Several preprocessing steps were performed before model training.

Data Cleaning:

- Converted TotalCharges column into numeric format
- Handled missing values using dropna()
- Removed unnecessary customerID column

Feature Engineering:

Three additional features were created to improve model performance:

1. ChargesPerMonth
    - Calculated average customer charges per month
2. SeniorWithNoSupport
    - Identified senior citizens without tech support services
3. TenureGroup
    - Grouped customers into tenure categories for better segmentation

Categorical variables were encoded using One-Hot Encoding because most categorical columns were nominal and did not contain ordinal relationships.

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4. Machine Learning Models Used

Three classification models were trained and evaluated:

1. Logistic Regression
2. Random Forest Classifier
3. Gradient Boosting Classifier

The dataset was split into 80% training data and 20% testing data using stratified sampling to preserve churn distribution.

StandardScaler was applied before training Logistic Regression to normalize numerical features.

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5. Model Evaluation

The models were evaluated using multiple classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

After comparison, Logistic Regression achieved the best overall performance with the highest ROC-AUC score and balanced classification metrics.

Best Model Performance:

- Accuracy: ~79%
- ROC-AUC Score: ~0.84

Logistic Regression was selected as the final model because it provided stable performance, better interpretability, and lower chances of overfitting compared to the other models.

Hyperparameter tuning was also performed using RandomizedSearchCV to further improve model performance.

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6. Customer Risk Segmentation

The best-performing model was used to predict churn probabilities for customers.

Customers were segmented into three risk categories:

- High Risk → Probability ≥ 0.70
- Medium Risk → Probability between 0.40 and 0.69
- Low Risk → Probability < 0.40

Key Observations:

- High-risk customers generally had low tenure and high monthly charges
- Most high-risk customers used month-to-month contracts
- Low-risk customers mostly had long-term yearly contracts
- Customers with higher tenure showed lower churn probability

This segmentation helps businesses identify customers who require immediate retention strategies.

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7. Visualizations & Insights

Several visualizations were created during the project to better understand churn behavior:

- Correlation Heatmap
- Model Performance Comparison Chart
- ROC Curve Comparison
- Feature Importance Chart
- Churn Rate by Contract Type
- Customer Risk Tier Distribution
- Tenure Distribution Analysis
- Interactive Plotly Scatter Plot

Important Insights:

1. Tenure was one of the strongest indicators of churn. Customers with lower tenure were more likely to leave the service.
2. Customers paying higher monthly charges had a higher probability of churning.
3. Contract type strongly influenced churn behavior. Month-to-month customers churned significantly more than customers with one-year or two-year contracts.
4. Long-term contracts improved customer retention and reduced churn probability.

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8. Business Recommendations

Based on the project findings, the following recommendations were identified:

Recommendation 1:

The company should focus heavily on customer retention during the first few months of subscription. New customers showed the highest churn probability, so onboarding experience, customer support, and promotional offers should be improved during the early stages.

Recommendation 2:

The company should encourage customers to move from month-to-month contracts to yearly plans using discounts, loyalty rewards, or bundled benefits. Long-term contracts were strongly associated with lower churn rates.

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9. Limitations & Future Improvements

Although the model performed well overall, there are still some limitations in the project.

The dataset mainly included customer demographics, subscription information, and billing details. However, it did not include customer satisfaction scores, complaint history, competitor pricing, or network quality information.

Future improvements could include:

- Using advanced models such as XGBoost or CatBoost
- Applying SMOTE for better class imbalance handling
- Building a real-time Streamlit or Power BI dashboard
- Performing deeper feature engineering using customer usage behavior
- Using larger and more updated datasets

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10. Conclusion

This project successfully built a complete machine learning pipeline for customer churn prediction and customer risk segmentation.

The analysis demonstrated that machine learning can effectively identify customers who are likely to churn and help businesses take proactive retention actions.

The final solution combined predictive modeling, risk segmentation, and business insights to create a practical and interpretable churn analysis system suitable for real-world telecom businesses.